Intermediate Domain Prototype Contrastive Adaptation for *Spartina alterniflora* Segmentation Using Multitemporal Remote Sensing Images

Boyu Zhao, Mengmeng Zhang, Wei Li, *Senior Member, IEEE*, Xiukai Song, Yunhao Gao, Yuxiang Zhang, and Junjie Wang

***Abstract*— As an invasive plant in wetlands, *Spartina alterniflora* (*S. alterniflora*) causes immeasurable damage to wetland ecosystems. Observing S.alterniflora using multitemporal remote sensing data helps us better understand its further development and facilitates effective containment of its invasion trend. However, inconsistent representation across remote sensing data from different time periods poses a challenge. Fortunately, the utilization of unsupervised domain adaptation (UDA) techniques helps in addressing such issues and enables the exploration of rich temporal dimension information in multitemporal remote sensing data, revealing the spatio-temporal distribution characteristics of S.alterniflora. However, existing UDA methods mostly focus on directly aligning the global or intraclass distribution representations across domains, which overlooks the issue of significant differences between extreme domains and lacks exploration of interclass relationships. To address these limitations, an intermediate domain prototype class-level learning network (IDPNet) is proposed. IDPNet utilizes dynamically generated intermediate domain (ID) features to construct class prototypes while incorporating interclass information into the prototype construction, achieving the class-centered distribution alignment for adaptation. Moreover, intermediate domain feature generation module (IFM) is employed in IDPNet to blend the latent representations from various domains and generate ID features in real time. Additionally, the hierarchical feature fusion module (HFM) is designed to enable IDPNet to learn more discriminative and robust spatio-temporal distribution features, thereby reducing the loss of information from patches. Experimental results on two cross-year multispectral datasets demonstrate that the proposed IDPNet outperforms several state-of-the-art UDA methods.**

***Index Terms*— Intermediate domain (ID) prototypical contrast adaptation, multitemporal remote sensing image segmentation, *Spartina alterniflora* (*S. alterniflora*), unsupervised domain adaptation (UDA).**

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I. Introduction

With the advancements in Earth observation systems, an increasing amount of wetland remote sensing data is now available for utilization [1], [2]. The segmentation of the invasive plant species, *Spartina alterniflora* (*S.alterniflora*) in wetland areas holds significant importance for effective invasion prevention and control. Among the available data sources, multispectral images (MSIs) offer high spatial resolution and extensive coverage, providing valuable information for ground observations [3]. Compared to MSIs, multitemporal remote sensing images provide continuous spatio-temporal distribution information, enabling accurate and timely capture of the invasion mechanisms of S.alterniflora [4]. Annotating remote sensing data, particularly multitemporal data, is time-consuming and labor-intensive due to the need for labeling images across different times, posing a challenge in extracting representative features efficiently. On the other hand, the majority of existing segmentation methods are designed for fixed scenarios where the training and test samples are independent and identically distributed. For example, Traditional methods include representation-based methods [5], [6] and least squares regression-based methods [7], [8], while deep learning methods encompass convolutional neural network (CNN)-based methods [9], [10] and graph convolutional network (GCN)-based methods [11]. Therefore, these methods cannot directly transfer to different temporal scenes in the application of multitemporal data segmentation, and they cannot test real-time acquired data without labels, which is the main challenge faced by cross-scene segmentation tasks. The training samples and testing samples are source domain (SD) and target domain (TD), respectively. The objective of this task is to train a model on SD and transfer the shared knowledge from SD to TD, enabling the model to adapt to different temporal scenes without the need for additional annotation costs.

The process of acquiring multitemporal and multispectral data is inevitably influenced by various factors, such as different seasons, acquisition times, and weather conditions [12], [13]. This leads to differences in spectral reflectance and conditional distributions of the same land cover classes between SD and TD [14]. Therefore, when directly applying SD for segmentation on TD, state-of-the-art methods still suffer from significant performance degradation due to domain shift issues [15], [16]. Fortunately, unsupervised domain adaptation

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(UDA) is a promising direction to address such problems. By exploring the correlations between SD and TD, UDA aims to reduce domain shift at the feature level [17], [18], and it has been widely applied in cross-scene segmentation tasks [19].

Existing UDA methods focus on discovering domain-invariant feature representations and lessening domain disparity, often by directly reducing the maximum mean difference (MMD) distance within the same subspace between two domains [20], [21]. Other methods [22], [23] rely on adversarial frameworks and a series of adversarial training losses to reduce the domain discrepancy between SD and TD. Like Cscada [24], combines contrastive learning with cross-structure domain adaptation and domain-specific batch normalization for effective domain transition. In this adversarial framework, a domain discriminator is trained to distinguish between SD and TD, while a feature extraction network is trained to generate domain-invariant features to deceive the domain discriminator, ultimately aligning the features between the two domains and making them indistinguishable.

Despite the significant progress made by metric-based and adversarial training-based methods in UDA, most of the efforts focus on aligning global feature distributions and pay less attention to the underlying structural relationships between classes. Some studies [25], [26] have shown that matching marginal features alone does not guarantee small expected error on TD, and aligning class conditional distributions is also necessary. As shown in Fig. 1(a), global features can align the two domains well, but some samples are incorrectly confused. Some UDA methods attempt to address this issue. In terms of improvement in distance metric methods, domain adaptation shape prior network (DASP) [27] encodes domain-invariant spatial relationships between segmentation categories, thereby aligning high-order statistics computed for the SD and TD to achieve domain adaptation. Zhu et al. [28] proposed the deep subdomain adaptive network (DSCN), which uses class labels as partition criteria and applies local MMD (LMMD) to align similar samples. In terms of improvement in adversarial frameworks, Luo et al. [29] proposed the class-level alignment network (CLAN), which purifies features and introduces class-level adversarial adaptation to achieve class-level feature alignment in semantic segmentation, reducing negative transfer phenomena. Similarly, Wang et al. [30] attempted to incorporate class information into feature alignment and proposed the fine-grained adversarial learning framework (FADA), merging class information into the discriminator to align features in a class-aware manner, thereby achieving better feature adaptation performance. Rangwani et al. [31] employs the smooth domain adversarial training approach (SDAT) to converge the challenging adversarial task loss to a smooth minimum, enabling effective model generalization across different scenes. However, these current class-level alignment methods overlook the consistency of the representation structure between SD and TD, meaning that these methods tend to bring similar class-level features closer while neglecting to push different class-level features apart, resulting in approximate class features on TD being projected into the same class. Therefore, considering only the intraclass distribution may not be sufficient to obtain well-performing class-level representations on TD [32]. As shown in Fig. 1(b), only with the push of interclass relationships can the features to be aligned be correctly projected, and relying solely on intraclass relationships cannot achieve this.

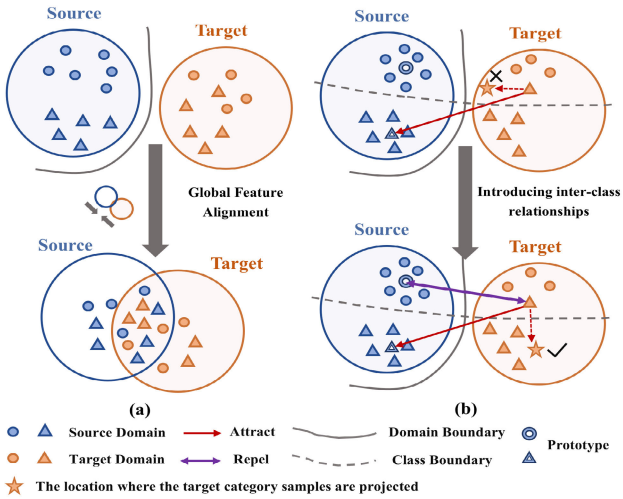


Fig. 1. Addresses the issues in traditional methods and introduces class-level relationship modeling. (a) Traditional global feature alignment strategy, which overlooks the alignment of class-level relationships. The top part of (b) illustrates the general class-level alignment process, while the bottom part depicts the alignment process with the introduction of interclass constraints. With the incorporation of interclass relationships, the adaptive features in the TD can appear at the correct position of the decision boundary, i.e., the location of ★.

By constructing class prototypes, it is possible to better describe the similarity or distance between other samples and that class, thus providing a better representation of class-level relationships. Jiang et al. [33] described class relationships by constructing prototypes and aligned TD pixels with their corresponding SD class through contrastive learning, while also pushing them further away from other prototypes. However, all the aforementioned UDA methods assume that the SD and TD are isolated from each other, without considering the presence of an intermediate state between them. This leads to difficulties in directly transferring knowledge between these isolated domains when the domain discrepancy is significant [34], [35]. Similarly, as shown in Fig. 2, prototype features constructed directly from two domains cannot accurately represent class-level information, and using these prototype features alone cannot effectively align the intraclass distribution and push apart interclass relationships. Moreover, due to the significant fragmentation in the distribution of S.alterniflora, direct feature extraction methods tend to miss important features of S.alterniflora [36], and such missing features can directly affect the accuracy of subsequent analysis and identification, resulting in incorrect ecological monitoring outcomes.

To address all the aforementioned issues, the generation of more prototypical class prototype features is pursued while considering the reduction of interdomain gaps. This is done with the ultimate aim of achieving cross-domain segmentation through contrastive learning. Through the exploration of bridging methods between SD and TD, the intermediate domain prototype-based class-level learning framework

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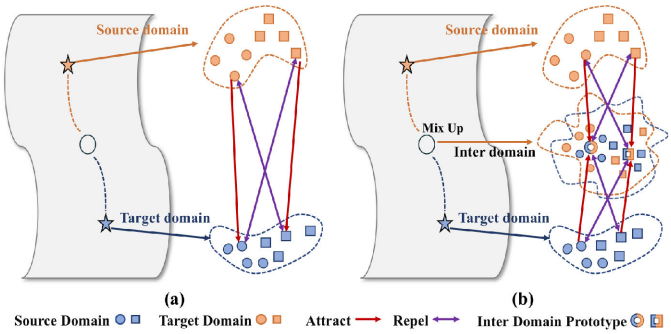


Fig. 2. Concept of an ID is introduced. (a) Directly modeling interclass relationships. (b) Modeling interclass relationships through the ID. As SD and TD exist within the same subspace, assuming the presence of some ID between these extreme domains, constructing class prototypes using appropriate IDs can reduce the influence of interdomain differences on prototype construction. Compared to directly modeling interclass relationships between the extreme domains (a), the knowledge from SD can be transferred more effectively to TD by leveraging the intermediate regions.

has been designed. In this framework, the generation of an intermediate domain (ID) is utilized to reduce the domain gap. After that, class prototype features are derived from the ID's features, contrastive learning is employed to reduce the distance between same-class features within the SD and TD. Therefore, the class discrimination is increased. At the feature and output levels of the network, it constructs ID prototype features in a multilevel manner and incorporates interclass information into the construction of prototypes, thereby achieving better modeling of class-level relationships. Furthermore, the cross-scene method [37] has shown that fusing information from various hierarchical levels during the feature extraction process aids in preventing the loss of fragmented details. Consequently, a multiscale hierarchical feature fusion module (HFM) has been devised, amalgamating data across scales. It enhances the model's detection of S.alterniflora's distribution and its adaptability and robustness in complex environments, ensuring precise and efficient feature extraction while preserving fragmented information. The main contributions of this work are summarized as follows.

1) To address the issues of insufficient class-level representation and large domain discrepancy, ID prototype class-level learning network (IDPNet) is proposed for ID prototype class-level learning. Specifically, we use dynamically generated fine-grained ID features as the core to smoothly bridge the SD and TD. We introduce constraints based on different class prototype characteristics and construct ID prototype features in a multilevel manner to enhance the class adaptation ability between the two domains.
2) Since the construction of prototype features requires multiple IDs as references to better represent domain-invariant information, we propose a plug-and-play ID feature generation module (IFM). This module can adaptively generate different scale coefficients to dynamically blend diverse hidden representations from the SD and TD, thereby mining more representative ID features.
3) To prevent the loss of fine-grained S.alterniflora information during feature extraction and simultaneously learn more discriminative and robust features, an HFM is designed. This module employs a hierarchical fusion approach to extract multiscale semantic features without reducing information loss.

The remainder of this article is organized as follows. Section II presents the details of the proposed IDPNet. Section III provides the experimental results together with the corresponding analysis and discussion. Finally, conclusions are drawn in Section IV.

II. Proposed IDPNet Framework

The framework of the proposed IDPNet is illustrated in Fig. 3, which consists of four parts: construction of ID features, construction and updating of class prototypes using ID features, adaptation of ID prototypes, and hierarchical feature fusion during the feature extraction process. Specifically, for given source images and target images, the features $\boldsymbol{F}_s$ and $\boldsymbol{F}_t$ of the two domains are obtained through a shared feature extractor $\mathcal{F}$. These features $\boldsymbol{F}_s$ and $\boldsymbol{F}_t$ are then input into the IFM, which generates the ID features $\boldsymbol{I}_F$ at the feature level. The generated ID features $\boldsymbol{I}_F$ are further fed into the classifier $\mathcal{C}$, which calculates the segmentation loss, to obtain the ID features $\boldsymbol{I}_O$ at the output level. After constructing prototypes at multiple levels using these two sets of ID features, pixels from both domains are contrasted with class-aware prototypes, directly modeling interclass constraints. In addition, during the training process, the HFM is placed after each layer of the feature encoder to extract multiscale high-level semantic information. This allows the model to learn more discriminative and robust class features, while avoiding the loss of smaller S.alterniflora patch information during the encoding process.

Assume the data from SD is $\boldsymbol{X}^s = \{x_i^s\}_{i=1}^{n_s} \in \mathbb{R}^d$ and the data from TD is $\boldsymbol{X}^t = \{x_i^t\}_{i=1}^{n_t} \in \mathbb{R}^d$. Since TD does not contain labeled data, the class label corresponding to SD is denoted as $\boldsymbol{Y}^s = \{y_i^s\}_{i=1}^{n_s} \in \mathbb{R}^d$. Here, d, n_s, and n_t represent the number of source samples, the number of target samples, and the dimensionality of the data, respectively.

A. Intermediate Domain Feature Generation Module

The purpose of the proposed IFM is to generate appropriate ID between SD and TD. By simultaneously training labeled SD and unlabeled TD, IFM dynamically generates ID in an adaptive manner. These ID smoothly bridge the gap between SD and TD, adapting to the differences between the two extreme domains. The prototypes generated through these ID are more representative of the classes, thereby constraining the encoder to align the distributions of SD and TD more effectively.

The structure of IFM is illustrated in Fig. 4. The IFM block consists of squeeze-and-excitation (SE) block [38], pooling layer, fully connected (FC) layer, multilayer perceptron (MLP), and softmax function. Specifically, the IFM block has two inputs, which are the features $\boldsymbol{F}_s$ extracted from SD by the feature extractor $\mathcal{F}$ and the features $\boldsymbol{F}_t$ extracted from TD (both $\boldsymbol{F}_s$ and $\boldsymbol{F}_t$ are the feature map $\in \mathbb{R}^{c\times h\times w}$). The feature map pair ($\boldsymbol{F}_s$, $\boldsymbol{F}_t$) is initially fed into two SE

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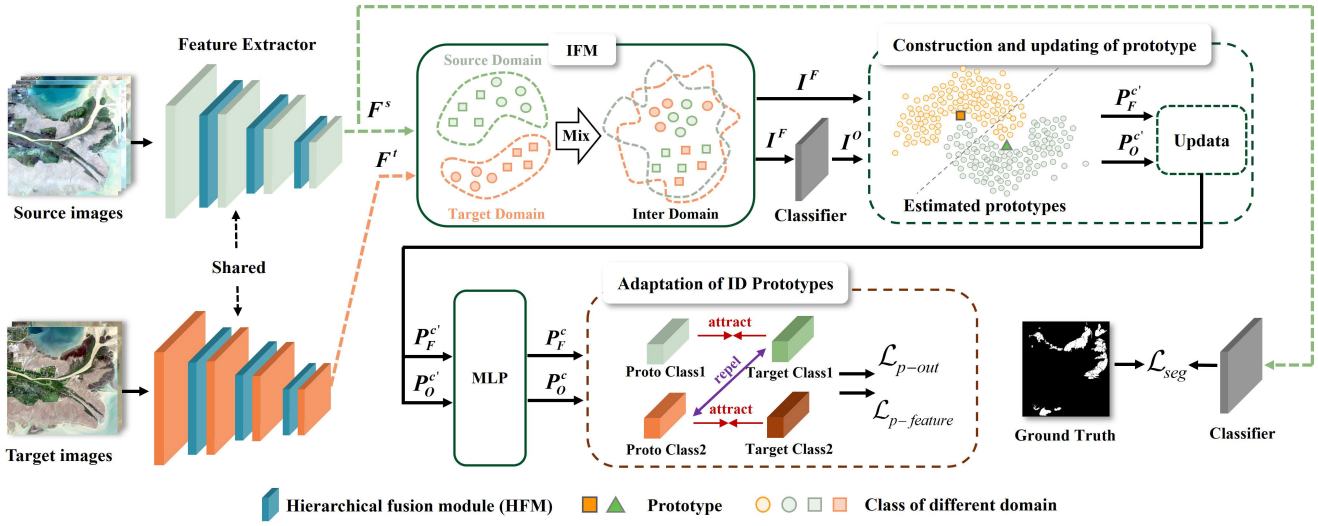


Fig. 3. Overall network architecture, which includes: 1) IFM; 2) construction and updating of prototypes; 3) adaptation of prototypes; and 4) HFM. The network performs ID prototypical adaptation at both the feature level and output level, updating and constraining the ID prototypes in a multilevel manner during the training process to enhance domain-invariant representation capability.

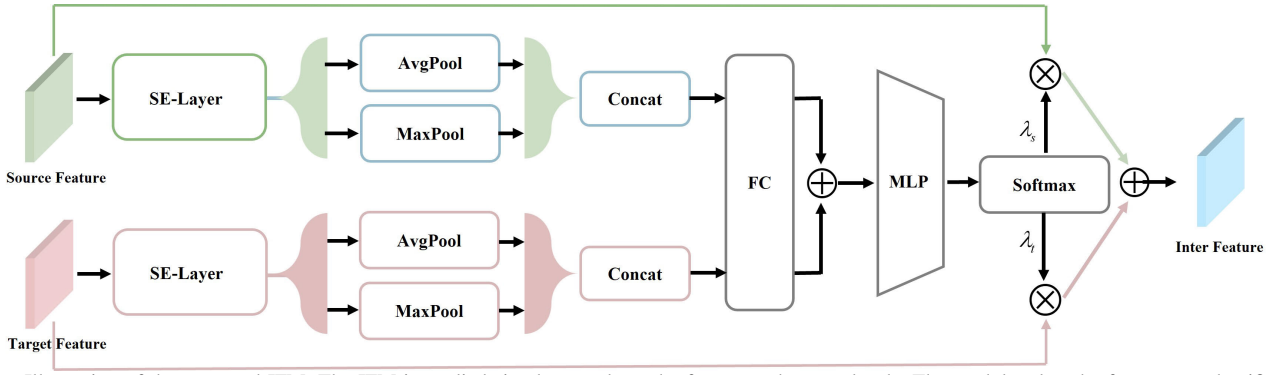


Fig. 4. Illustration of the proposed IFM. The IFM is applied simultaneously at the feature and output levels. The module takes the features or classification results extracted from SD and TD as inputs and generates two domain factors, namely λ_s and λ_t. By using these two domain factors, the hidden representations of SD and TD are mixed to dynamically and multiscale generate ID in real-time.

modules separately, aiming to preserve the original features while learning the relationships between different channels. Subsequently, the feature pairs $(\boldsymbol{G}_s, \boldsymbol{G}_t)$ obtained from the SE modules undergo average pooling and max pooling operations, resulting in $1 \times 1 \times c$-dimensional features. This is done to generate domain factors that are more sensitive to feature position variations in the input. The aforementioned steps can be represented as follows:

$$\begin{aligned} \boldsymbol{G}_s &= \mathrm{SE-layers}(\boldsymbol{F}_s), \boldsymbol{G}_t = \mathrm{SE-layers}(\boldsymbol{F}_t) \\ \boldsymbol{G}_s^{\mathrm{avg}} &= \mathrm{Avgpool}(\boldsymbol{G}_s), \boldsymbol{G}_t^{\mathrm{avg}} = \mathrm{Avgpool}(\boldsymbol{G}_t) \\ \boldsymbol{G}_s^{\mathrm{max}} &= \mathrm{Maxpool}(\boldsymbol{G}_s), \boldsymbol{G}_t^{\mathrm{max}} = \mathrm{Maxpool}(\boldsymbol{G}_t). \end{aligned} \tag{1}$$

Next, the average features and max features of each domain are concatenated and fed into the FC layer. The output feature vectors of FC are merged using elementwise summation and passed through an MLP to obtain the domain factor $\lambda = [\lambda_s, \lambda_t] \in \mathbb{R}^2$. The process to obtain the domain factor is as follows:

$$\begin{aligned} \boldsymbol{G}_s^2 &= \mathrm{concat}(\boldsymbol{G}_s^{\mathrm{avg}}, \boldsymbol{G}_s^{\mathrm{max}}) \\ \boldsymbol{G}_t^2 &= \mathrm{concat}(\boldsymbol{G}_t^{\mathrm{avg}}, \boldsymbol{G}_t^{\mathrm{max}}) \\ \lambda &= \sigma(\mathrm{MLP}(\mathrm{FC}(\boldsymbol{G}_s^2) + \mathrm{FC}(\boldsymbol{G}_t^2))) \end{aligned} \tag{2}$$

where $\sigma(\bullet)$ is the softmax function, $\boldsymbol{G}_s^2 \in \mathbb{R}^{1\times1\times2c}$ and $\boldsymbol{G}_t^2 \in \mathbb{R}^{1\times1\times2c}$ represent the concatenated features of SD and TD, respectively. Then, the operation of generating ID representations by mixing the source and target representations with two domain factors is as follows:

$$\boldsymbol{I}_F = \lambda_s \times \boldsymbol{F}_s + \lambda_t \times \boldsymbol{F}_t \tag{3}$$

where λ_s and λ_t are the two domain factors for the SD and TD, respectively ($[\lambda_s, \lambda_t] = \lambda$). The $\boldsymbol{I}_F$ represents the ID at the feature level. Additionally, it is necessary to synthesize the ID representation at the output level to construct ID class prototypes in a multiscale manner. This ultimately enables multiscale prototypical adaptation. The operation steps for generating the ID representation at the output level are

$$\boldsymbol{I}_O = \mathrm{Classifier}(\boldsymbol{I}_F) \tag{4}$$

where Classifier is shared with the classifier used for calculating the segmentation loss. The output dimension of $\boldsymbol{I}_O$ is $C \times h \times w$, where C represents the number of classes. At this stage, the ID has been constructed. Next, class prototypes will be built using the ID features.

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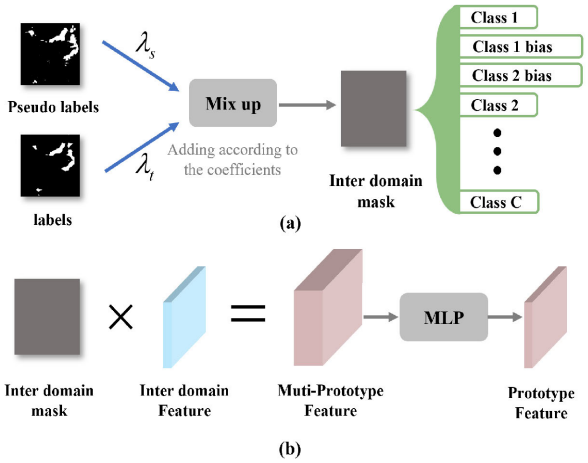


Fig. 5. Illustrates the process of prototype feature construction. (a) Building ID labels using the labels of SD, pseudolabels of TD, and domain factors to adapt the ID features. Assuming there are C classes in the images, the number of classes in the ID labels are C^2. (b) Process of prototype construction using the ID labels and their corresponding ID features. The multiprototype features are dimensionally reduced using an MLP to obtain the prototype features.

B. Construction and Updating of ID Class Prototypes

As shown in Fig. 5, calculating class prototypes requires obtaining labels from both domains. The labels of SD are known, while the labels of TD are unknown. To obtain the pseudolabels for TD, a method is employed that involves selecting high-confidence samples based on a confidence threshold [39], [40]. This method allows for the real-time construction and utilization of pseudolabels for TD. Specifically, the output of the classifier is first passed through a softmax layer to calculate the confidence of the cth class and sort them in descending order. This yields a confidence set for the class, $a^c = [a_1^c, a_2^c, \ldots, a_{l_c}^c]$, where the length of the confidence set for the cth class, l_c, is computed using the following formula:

$$l_c = \sum_{n=1}^{n_t} \sum_{i=1}^{H} \sum_{j=1}^{W} \mathbb{1}[\boldsymbol{Y}_{n,i,j}^{tp} = c] \tag{5}$$

where $\boldsymbol{Y}^{tp} \in \mathbb{L}^{H\times W}, \mathbb{L} = \{1, 2, \ldots, C\}$ represents the unmasked predicted pseudolabel of TD. H and W denote the height and width of the pseudolabel. Next, the class threshold for the cth class in the pseudolabels is obtained by taking the pixel threshold at the ηth percentile of the predicted confidence set a^c. The portion below this threshold is removed using a mask. After performing the aforementioned operations on each class of $\boldsymbol{Y}^{tp}$, the masked pseudolabels $\boldsymbol{Y}^t$ are obtained. With the known labels $\boldsymbol{Y}^s$ for SD, the domain factors obtained from IFM are applied for the synthesis of the ID labels $\boldsymbol{Y}^I$. The synthesis operation is as follows:

$$\boldsymbol{Y}^I = \lambda_s \times \boldsymbol{Y}^s + \lambda_t \times \boldsymbol{Y}^t. \tag{6}$$

The ID labels $\boldsymbol{Y}^I$ consist of C^2 classes, which means that between the first class and the second class, for example, there may exist four classes: the first class and second class shared by both SD and TD, the first class leaning toward SD, the second class leaning toward TD (or the second class leaning toward SD, the first class leaning toward TD). The specific classes depend on the magnitude of the domain factors, but the number of classes is the square of the original number of classes. The above methods enable the synthesis of ID labels with richer class information. This label not only contains more fine-grained class information but also provides support for the model to better understand the differences between domains. With the help of these labels, it becomes possible to synthesize ID prototypes that carry more comprehensive class information. By setting n_s to be equal to n_t for SD and TD, respectively, the number of ID images n_i becomes equal to n_s and n_t. In this case, the class-aware prototypes $\boldsymbol{P}_F^{c'}$ and $\boldsymbol{P}_O^{c'}$ can be initialized

$$\begin{aligned}\boldsymbol{P}_F^{c'} &= \frac{\sum_{n=1}^{n_i}\sum_{i=1}^{H}\sum_{j=1}^{W} I_F^{n,i,j}\,\mathbb{1}[Y_{n,i,j}^I = c']}{\sum_{n=1}^{n_i}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbb{1}[Y_{n,i,j}^I = c']} \\ \boldsymbol{P}_O^{c'} &= \frac{\sum_{n=1}^{n_i}\sum_{i=1}^{H}\sum_{j=1}^{W} I_O^{n,i,j}\,\mathbb{1}[Y_{n,i,j}^I = c']}{\sum_{n=1}^{n_i}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbb{1}[Y_{n,i,j}^I = c']}\end{aligned} \tag{7}$$

whereas $\boldsymbol{I}_F \in \mathbb{R}^d$ represents the d-dimensional ID feature vector obtained through IFM synthesis at the feature level, and $\boldsymbol{I}_O \in \mathbb{R}^C$ represents the C-dimensional ID feature vector obtained through IFM synthesis at the output level. The value of d is determined based on the network depth, typically set to 2048, while C represents the number of classes. The variable c' in the equation represents the class of the ID label, and the total number of possible classes is C^2. The computed class prototype feature $\boldsymbol{P}_F^{c'} \in \mathbb{R}^{C^2\times d}$, while $\boldsymbol{P}_O^{c'} \in \mathbb{R}^{C^2\times C}$. To enhance the domain-invariant representational ability of the prototypes and incorporate class-related information from different domains during the training process, a scheme of updating the prototypes based on the strict statistical mean of global data [32] is adopted

$$\boldsymbol{P}_F^{c'} \leftarrow \frac{\boldsymbol{P}_F^{c'} n^{c'} + \widetilde{\boldsymbol{P}}_F^{c'} \widetilde{n}^{c'}}{n^{c'} + \widetilde{n}^{c'}}, \quad \boldsymbol{P}_O^{c'} \leftarrow \frac{\boldsymbol{P}_O^{c'} n^{c'} + \widetilde{\boldsymbol{P}}_O^{c'} \widetilde{n}^{c'}}{n^{c'} + \widetilde{n}^{c'}} \tag{8}$$

where c' represents the class in the ID label $\boldsymbol{Y}^I$, $n^{c'}$ represents the accumulated number of pixels belonging to class c' until the last update, $\widetilde{\boldsymbol{P}}_F^{c'}$ and $\widetilde{\boldsymbol{P}}_O^{c'}$ represents the online estimated prototypes for class c', and $\widetilde{n}^{c'}$ represents the total number of pixels belonging to class c' from a newly appended mini-batch during training.

After the updates to $\boldsymbol{P}_F^{c'}$ and $\boldsymbol{P}_O^{c'}$, in order to integrate the fine-grained information of prototypes into a single class prototype and enhance the representation capability of class prototypes, as well as to facilitate interclass modeling, the class dimension C^2 of prototype features needs to be mapped to C, the desired number of classes. To achieve this, as shown in Fig. 5(b), a learnable MLP module is employed to perform the mapping on the class dimension. By compressing the class dimension of fine-grained prototypes, instead of directly constructing class prototypes, class prototypes with stronger class representation capabilities can be created. The operations are shown as follows:

$$\boldsymbol{P}_F^c = \mathrm{MLP}(\boldsymbol{P}_F^{c'}),\ \boldsymbol{P}_O^c = \mathrm{MLP}(\boldsymbol{P}_O^{c'}). \tag{9}$$

The prototype features in the equation $\boldsymbol{P}_F^c \in \mathbb{R}^{C\times d}$, while $\boldsymbol{P}_O^c \in \mathbb{R}^{C\times C}$. The ID prototypes can be regarded

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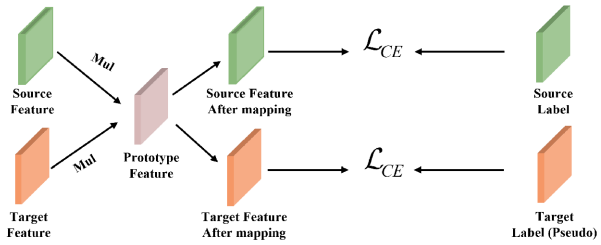


Fig. 6. Schematic illustration of the process of prototype adaptation. Through the cross-entropy function, the distances between the features of SD and TD and the prototypes constructed by ID for the same class are simultaneously reduced, while the distances between prototypes of different classes are increased.

as approximate representation centers for various classes. Next, a comparative adaptation is performed using the class prototypes.

C. Adaptation of ID Prototypes

The purpose of prototype domain adaptation is to make the features of the same classes in the source and target domains similar, while keeping the features of different classes far apart, thus utilizing the intraclass and interclass relationships, the specific implementation is shown in Fig. 6. After the images from SD and TD are fed into the shared feature extractor $\mathcal{F}$, the corresponding feature maps $\boldsymbol{F}_s$ and $\boldsymbol{F}_t$ are obtained at the feature level. After being input into the shared classifier $\mathcal{C}$, the output features $\boldsymbol{O}_s$ and $\boldsymbol{O}_t$ are obtained at the output level. With the updated $\boldsymbol{P}_F^c$ and $\boldsymbol{P}_O^c$, the similarity between the features and the prototypes of the ID classes can be calculated through mapping

$$\begin{aligned}
\boldsymbol{P}(f)_{s\to I}^{n,i,j,c} &= \frac{\exp(\boldsymbol{P}_F^c \times \boldsymbol{F}_s^{n,i,j})}{\sum_{c=1}^{C}\exp(\boldsymbol{P}_F^c \times \boldsymbol{F}_s^{n,i,j})} \\
\boldsymbol{P}(o)_{s\to I}^{n,i,j,c} &= \frac{\exp(\boldsymbol{P}_O^c \times \boldsymbol{O}_s^{n,i,j})}{\sum_{c=1}^{C}\exp(\boldsymbol{P}_O^c \times \boldsymbol{O}_s^{n,i,j})} \\
\boldsymbol{P}(f)_{t\to I}^{n,i,j,c} &= \frac{\exp(\boldsymbol{P}_F^c \times \boldsymbol{F}_t^{n,i,j})}{\sum_{c=1}^{C}\exp(\boldsymbol{P}_F^c \times \boldsymbol{F}_t^{n,i,j})} \\
\boldsymbol{P}(o)_{t\to I}^{n,i,j,c} &= \frac{\exp(\boldsymbol{P}_O^c \times \boldsymbol{O}_t^{n,i,j})}{\sum_{c=1}^{C}\exp(\boldsymbol{P}_O^c \times \boldsymbol{O}_t^{n,i,j})}.
\end{aligned} \tag{10}$$

$\boldsymbol{P}(f)_{s\to I}^{n,i,j,c}$ and $\boldsymbol{P}(f)_{t\to I}^{n,i,j,c}$ refer to the similarity between the features at the feature level and the output class prototype $\boldsymbol{P}_F^c$ constructed in ID. $\boldsymbol{P}(o)_{s\to I}^{n,i,j,c}$ and $\boldsymbol{P}(o)_{t\to I}^{n,i,j,c}$ refer to the similarity between the features at the output level and the class prototype $\boldsymbol{P}_O^c$ constructed in ID. After obtaining the similarity, the cross-entropy loss is computed separately based on the labels $\boldsymbol{Y}^s$ and $\boldsymbol{Y}_t$ on SD and TD, respectively,

$$\begin{aligned}
\mathcal{L}(f)_n^{s\to I} &= -\sum_{c=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W} y_{n,i,j,c}^s \log \boldsymbol{P}(f)_{s\to I}^{n,i,j,c} \\
\mathcal{L}(o)_n^{s\to I} &= -\sum_{c=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W} y_{n,i,j,c}^s \log \boldsymbol{P}(o)_{s\to I}^{n,i,j,c} \\
\mathcal{L}(f)_n^{t\to I} &= -\sum_{c=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W} y_{n,i,j,c}^t \log \boldsymbol{P}(f)_{t\to I}^{n,i,j,c}
\end{aligned}$$

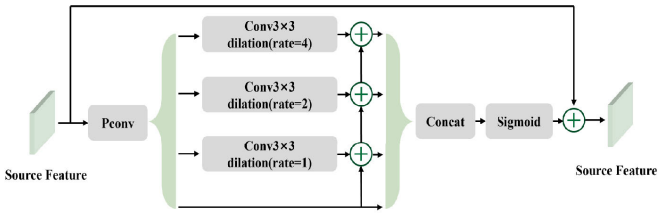


Fig. 7. Structure of HFM. By fusing multilevel features, the model's ability to extract fine-grained information about the invasive species S.alterniflora is enhanced, thereby improving the model's discriminative power and robustness.

$$\mathcal{L}(o)_n^{t\to I} = -\sum_{c=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W} y_{n,i,j,c}^t log \boldsymbol{P}(o)_{t\to I}^{n,i,j,c}. \tag{11}$$

Based on the aforementioned loss functions, the prototype adaptation losses, $\mathcal{L}_{\text{proto-}F}$ at the feature level and $\mathcal{L}_{\text{proto-}O}$ at the output level can be obtained, respectively,

$$\begin{aligned}
\mathcal{L}_{\text{proto-}F} &= \sum_{n=1}^{n_s}\mathcal{L}(f)_n^{s\to I} + \sum_{n=1}^{n_t}\mathcal{L}(f)_n^{t\to I} \\
\mathcal{L}_{\text{proto-}O} &= \sum_{n=1}^{n_s}\mathcal{L}(o)_n^{s\to I} + \sum_{n=1}^{n_t}\mathcal{L}(o)_n^{t\to I}.
\end{aligned} \tag{12}$$

By optimizing these cross-entropy loss functions, it is evident that the features at positions i and j become similar to their corresponding class prototypes, while the distance between them and other class prototype features is increased, thereby achieving feature alignment. Additionally, since the prototypes are continuously updated and iterated, the features of SD and TD are continuously aligned with the prototypes constructed by the ID generated from SD and TD. Through iterative training, the feature distributions of the three domains gradually become consistent, ultimately achieving cross-domain alignment.

Based on the above loss function, IDPNet combines the supervised loss $\mathcal{L}_{\text{seg}}$ on the source samples and the multiscale prototype adaptation losses $\mathcal{L}_{\text{proto-}F}$ and $\mathcal{L}_{\text{proto-}O}$ on both the source and target samples during the training process. The final optimization problem can be formulated as follows:

$$\begin{aligned}
\mathcal{L}(\boldsymbol{X}_s, \boldsymbol{X}_t, \boldsymbol{Y}_s) = \mathcal{L}_{\text{seg}}(\boldsymbol{X}_s, \boldsymbol{Y}_s) + \lambda_1\mathcal{L}_{\text{proto-}F} \\
+ \lambda_2\mathcal{L}_{\text{proto-O}}
\end{aligned} \tag{13}$$

where λ_1 and λ_2 are the regularization parameters controlling the different contributions of prototype adaptation losses at the feature and output levels.

D. Hierarchical Feature Fusion Module

The structure of the HFM is illustrated in Fig. 7. This module is embedded at each stage of the backbone network. Compared to directly using the backbone network to extract features from SD and TD, the HFM empowers the encoder with multiscale feature extraction capabilities, enabling it to capture finer details of S.alterniflora and ensure that the information of S.alterniflora is preserved. Specifically, the module first reduces the channel dimension to one-fourth of its original size using Pointwise Convolution [41]. This reduces the number of parameters for subsequent computations while facilitating the combination of multilevel features to prevent

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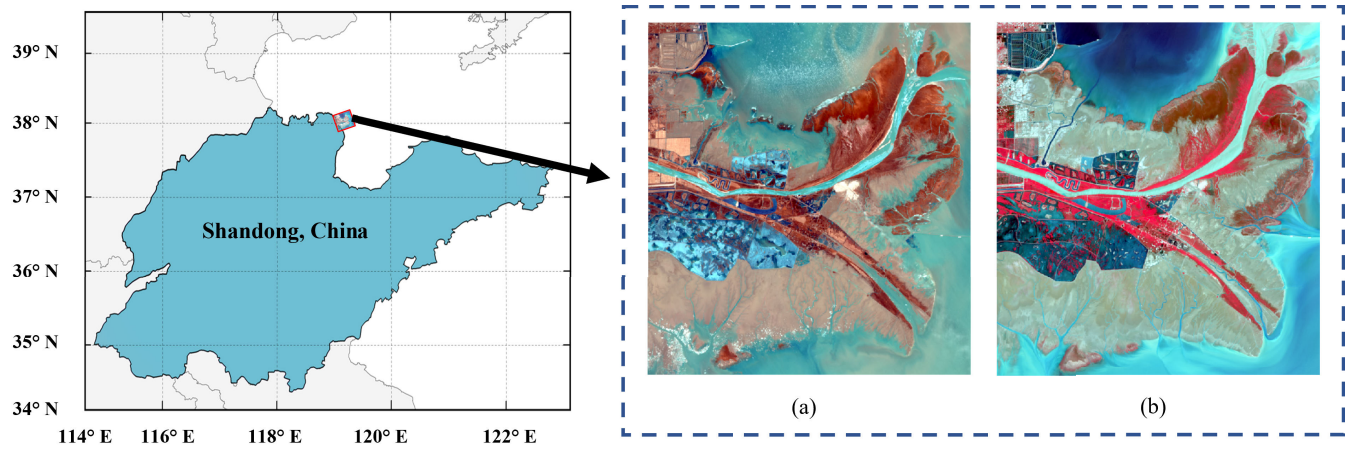


Fig. 8. Dataset of the Yellow River Estuary wetlands. (a) Pseudocolor image of the Yellow River Estuary region in winter. (b) Pseudocolor image of the Yellow River Estuary region in summer.

information loss. Then, dilation convolutions [42] are performed with dilation factors of 1, 2, and 4, respectively. This enlarges the receptive field and extracts contextual information at multiple scales without losing information or increasing parameters. In order to mitigate the grid artifacts caused by the increased sampling interval in dilated convolutions, while preserving the extracted contextual information, the design of the HFM draws inspiration from the hierarchical feature fusion method proposed in efficient spatial pyramid of dilated convolutions for semantic segmentation version 2 (ESPNetV2) [43]. Starting from the feature map generated by the smallest dilated convolution kernel, the HFM progressively accumulates features in a hierarchical manner. This approach ensures the preservation of a larger effective receptive field while resolving the grid artifacts [44]. Specifically, the results of the Pointwise Convolution and the features obtained from three dilated convolutions are sequentially summed up in the order of increasing receptive fields, compensating for the information loss due to the discontinuous sampling. Finally, the features at different scales are concatenated along the channel dimension.

III. Experimental Results and Discussion

A. Dataset Description and Preprocessing

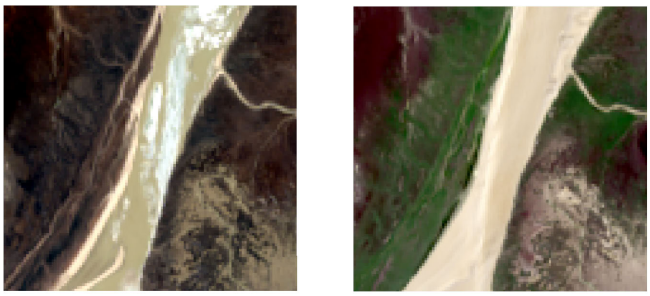
To validate the cross-temporal recognition capability of IDPNet for S.alterniflora in remote sensing images, the dataset used in this study is distributed in the northeastern region of the Yellow River Delta in Shandong Province, China, where S.alterniflora erosion is relatively severe, as shown in Fig. 8. Currently, the scientific management of S.alterniflora in the wetlands of the Yellow River Estuary faces significant challenges due to the lack of remote sensing interpretation data for future years. In order to analyze the invasion mechanisms and directional changes of S. alterniflora, and to better protect and develop the wetlands of the Yellow River Estuary, experiments are designed to interpret multitemporal remote sensing images, i.e., using existing remote sensing data to predict future remote sensing data. The purpose of the experiments is to predict the invasion of S.alterniflora in future years, in order to effectively address the ecological hazards caused by the invasion and expansion of S.alterniflora in wetlands. By predicting the invasion areas of S.alterniflora in future remote sensing images, it is not only crucial for the ecological protection and development of coastal wetlands but also provides support for the analysis of long-term wetland geographical and environmental changes.

TABLE I
Overview of the Combined Landsat Data Used in This Study (Date Format: Year/Month/Day)

Period	Sensor	Imaging Time	Number of Bands	Proportion
2018	Landsat 8 OLI	2018/03/25	6	7.09%
		2018/09/17	6	7.23%
2019	Landsat 8 OLI	2019/02/24	6	7.52%
		2019/08/19	6	8.08%
2020	Landsat 8 OLI	2020/03/14	6	8.33%
		2020/07/20	6	8.35%
2021	Landsat 8 OLI	2021/01/28	6	9.50%
		2021/06/21	6	9.90%

The construction of the dataset involved the use of eight original Landsat images from the years 2018, 2019, 2020, and 2021, spanning eight different time periods (as listed in Table I). The selected remote sensing images were concentrated in the months of January to April and July to November, representing two quarters with significant spectral differences in S.alterniflora, as illustrated in Fig. 9. This selection was made to ensure that the trained model would maintain good interpretative accuracy for S.alterniflora even during seasonal variations. Additionally, efforts were made to avoid cloud cover obstructing the study area when selecting the remote sensing images. In this study, six original bands of Landsat 8 OLI, namely Red, Green, Blue, near

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(a) (b)

Fig. 9. Image depicts a portion of the Yellow River Estuary wetlands captured in true-color remote sensing images. (a) Image taken in winter. (b) Image taken in summer. From the image, it can be observed that the seasonal variations have a significant impact on S.alterniflora. In winter, S.alterniflora appears reddish-brown, while in summer, it appears green.

TABLE II
NUMBER OF IMAGES CORRESPONDING TO EACH EXPERIMENT

Year	Numbers of patches (steps=64)	Experiment1		Experiment2	
		Source	Target	Source	Target
2018-1	2300	✓		✓	
2018-2	2300	✓		✓	
2019-1	2300	✓		✓	
2019-2	2300	✓		✓	
2020-1	2300	✓		✓	
2020-2	2300	✓		✓	
2021-1	156		✓		
2021-2	156				✓
Total Num		13880	156	13880	156

infrared (NIR), shortwave infrared 1 (SWIR-1), and shortwave infrared 2 (SWIR-2), were selected. The panchromatic band was excluded from the analysis. For this study, ENVI 5.3 software was selected to perform a series of preprocessing on the collected Landsat8 OLI data to reduce the effect of data noise on the overall experimental results. First, the numbers were converted into radiance by radiometric correction method, and the problem of data radiometric errors caused by atmospheric scattering and absorption was eliminated by the quick atmospheric correction (QUAC) method. Finally, the processed images are cropped and the bands are combined.

Due to factors such as different years, weather conditions, and quarters, there are spectral shifts that occur. Using remote sensing data from other years directly to segment the remote sensing data of the target year would lead to spectral discrepancies and result in poor accuracy in results. In order to improve the interpretative accuracy of the target year and quarter remote sensing data and achieve cross-temporal scene prediction of S.alterniflora throughout the entire quarter, this study selects two quarters of remote sensing images from 2018 to 2020 as SD and remote sensing images from two quarters of 2021 as TD to validate the effectiveness of the IDPNet model. The remote sensing images of each year and their corresponding vector labeled images are segmented, resulting in numerous image patches of size 64 × 64 pixels. To ensure training effectiveness, a segmentation stride of 32 is set for the SD years, while a stride of 64 is set for the TD year. Since separate experiments are conducted for the first and second half of 2021, two sets of different SD and TD image pairs are obtained, as listed in Table II. In the investigated images, S.alterniflora is assigned to class 1, while all other land types are considered as background.

B. Settings of Experiments

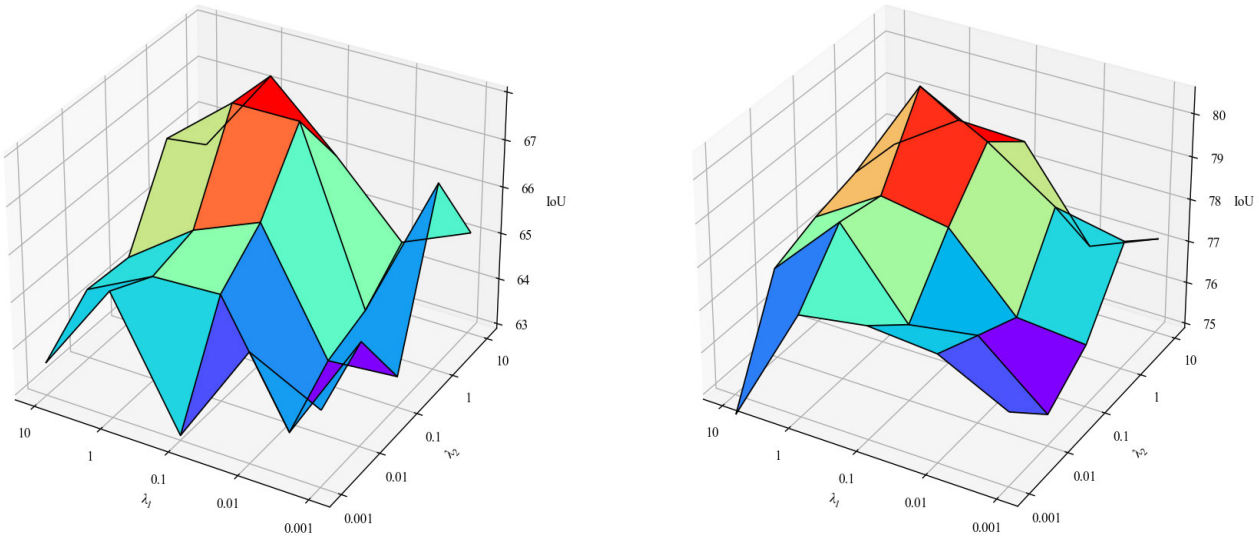
In evaluating the experimental results, intersection over union (IoU), which is the most commonly used metric for semantic segmentation, is selected as the primary evaluation metric to assess the performance of all models. Furthermore, as the experiment involves a binary problem, precision and recall are employed to evaluate the models' performance from both predictive and actual perspectives, respectively. Additionally, F1 scores are utilized to provide a scientifically valid measure of each metric in binary classification. The evaluation metrics are calculated as demonstrated below

$$\begin{cases} \text{IoU} = \dfrac{\text{TP}}{\text{FN}+\text{FP}+\text{TP}} \\ \text{Precision} = \dfrac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Recall} = \dfrac{\text{TP}}{\text{TP}+\text{FN}} \\ F1-\text{score} = \dfrac{2\text{TP}}{\text{FP}+2\text{TP}+\text{FN}} \end{cases} \tag{14}$$

where the pixel number that correctly predicts the target represents true positives (TP). False positives (FP) indicate the number of pixels from other classes that predict the target incorrectly. False negatives (FN) are the pixel number that predicts the target incorrectly. True negatives (TN) indicate the correct prediction of other pixels.

Some advanced and authoritative algorithms are used to compare with IDPNet in order to validate the superiority of the proposed model. These algorithms include AdaptSeg [22], ADVENT [23], MaxSquare [45], DSCN [28], CLAN [29], FADA [30], DALN [46], ProCA [33], DASP [27], Cscada [24], and SDAT [31]. These networks used for training and testing are referred to as SD and TD, as listed in Table II. Following the majority of previous work, all experiments are conducted using the DeepLab-v2 framework [47] as the segmentation model for fair comparison. To ensure fairness, IDPNet is compared with other networks in two separate experiments. One experiment utilizes the same encoder, ResNet-101 [48], as the comparative network, while the other experiment incorporates the designed HFM into the encoder. All models are pretrained on ImageNet [49] and employed an upsampling layer to obtain final per-pixel predictions with the same image size as the input. In addition, all methods incorporate supervised training for the segmentation models in the labeled SD, aiming to minimize the loss between the predicted values p_n^s and the ground-truth label map Y^s, which is annotated with C class labels. For a given image X^s, the model training is optimized by minimizing the loss function. In this study, C is equal

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(a) (b)

Fig. 10. Parameter tuning of λ_1 and λ_2 for the proposed IDPNet using all the two experimental data. (a) 2021-1. (b) 2021-2.

TABLE III
SCORE (%) OF IDPNET IN 2021-1 AFTER ABLATION OF EACH STRUCTURE

Model Structure				2021-1			
Source only	Proto-Feature	Proto-Output	HFM	IoU	Precision	Recall	Fscore
✓				56.2	87.34	60.86	71.62
✓	✓			64.06	87.86	70.28	78.08
✓		✓		65.34	83.92	74.84	78.98
✓	✓	✓		66.74	86.16	74.76	79.98
✓	✓	✓	✓	67.36	84.32	77.04	80.44

TABLE IV
SCORE (%) OF IDPNET IN 2021-2 AFTER ABLATION OF EACH STRUCTURE

Model Structure				2021-2			
Source only	Proto-Feature	Proto-Output	HFM	IoU	Precision	Recall	Fscore
✓				72.92	87.66	81.3	84.32
✓	✓			77.84	89.82	85.42	87.54
✓		✓		78.42	89.42	86.5	87.94
✓	✓	✓		79.24	89.84	87.04	88.4
✓	✓	✓	✓	80.18	90.44	87.6	88.98

to 2, with one class representing S.alterniflora and the other class representing the background. The standard cross-entropy loss is used in this study and can be represented as follows:

$$\mathcal{L}_{\text{seg}} = \sum_{n=1}^{n_s} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{c=1}^{C} y^s_{n,i,j,c} \log(p^s_{n,i,j,c}) \tag{15}$$

where n_s is the number of SD images, H and W denote the height and the width of an image, and $y^s_n \in 0, 1^{H\times W\times C}$ is the one-hot representation of the ground-truth label Y^s_n.

Experiments are all conducted on a computer equipped with a GeForce RTX 3090 GPU with 24 GB of RAM. The optimizer function uses the stochastic gradient descent (SGD) function [50], and the learning rate is set to 0.002. The final output of all networks is a two-channel tensor, and the value of each pixel in the tensor represents whether the pixel is S.alterniflora or background. In addition, the batch size of network training is set to 8 and the training epoch is set to 50. Finally, to ensure the fairness of the experimental results, the results of each network are averaged over ten times.

C. Parameter Tuning and Ablation Study

In the proposed IDPNet, adjustable hyperparameters include the regularization parameters λ_1 and λ_2, which, respectively, control the prototype contrastive adaptation contributions at the output level and output level. The selection of optimal hyperparameters utilizes tenfold cross validation, with the range set as $\{1e-3, 1e-2, 1e-1, 1e+0, 1e+1\}$. Fig. 10

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TABLE V
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT NETWORKS IN 2021-1

Network	Experiment Results			
	IoU	Precision	Recall	F1-score
Baseline	56.2	**87.34**	60.86	71.62
AdaptSeg [22]	59.9	85.36	66.12	74.46
ADVENT [23]	59.32	85.36	66.12	74.46
Maxsquare [45]	62.38	86.1	69.42	76.8
DSCN [28]	58.78	84.26	66.2	74.04
CLAN [29]	63.56	82.9	73.34	77.68
FADA [30]	64.3	82.72	74.34	78.26
DALN [46]	61.64	75.16	77.46	76.16
ProCA [33]	65.4	83.42	75.2	79.06
DASP [27]	60.3	81.68	70.26	75.22
Cscada [24]	61.74	81.52	71.74	76.26
SDAT [31]	63	86.66	69.84	77.2
IDPNet (only proto)	66.74	86.16	74.76	79.98
IDPNet (all)	**67.36**	84.32	**77.04**	**80.44**

TABLE VI
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT NETWORKS IN 2021-2

Network	Experiment Results			
	IoU	Precision	Recall	F1-score
Baseline	72.92	87.66	81.3	84.32
AdaptSeg [22]	74.5	90.86	80.58	85.38
ADVENT [23]	73.58	89.02	80.98	84.76
Maxsquare [45]	75.46	88.46	83.72	86.02
DSCN [28]	75	88.96	82.72	85.7
CLAN [29]	75.34	90.84	81.56	85.9
FADA [30]	77.52	87.48	87.24	87.34
DALN [46]	76.46	86.66	86.74	86.66
ProCA [33]	77.76	89.54	85.6	87.52
DASP [27]	75.46	91.22	81.44	86.02
Cscada [24]	76.38	**92.38**	81.8	86.8
SDAT [31]	77.5	91.42	83.54	87.3
IDPNet (only proto)	79.24	89.24	87.04	88.4
IDPNet (all)	**80.18**	90.44	**87.6**	**88.98**

shows the changing trend of the IoU of IDPNet in two different test data. By observing the trend of hyperparameter tuning, both parameters λ_1 and λ_2 are found to perform well within the range of $1e+0$ to $1e+1$, indicating that IDPNet achieves good segmentation performance. When λ_1 or λ_2 is reduced to a small value, even if the other parameter is large, the overall segmentation performance is relatively poor. This confirms that IDPNet is highly sensitive to both parameters λ_1 and λ_2, and multilevel prototype adaptation significantly improves the overall network performance. To achieve a balanced impact of prototype adaptation at both the output level and feature level and to reduce the pressure of hyperparameter tuning, for the data from the two testing quarters of 2021, λ_1 and λ_2 are, respectively, set to $1e+0$.

After selecting the hyperparameters, the contributions of the important components of IDPNet are being evaluated. IDPNet mainly includes two level prototype adaptation constraints at the feature level and output level, along with an HFM that enhances the encoder's feature extraction capability. Ablation analysis is being conducted on two different test datasets collected in different seasons, which exhibit significant differences. This demonstrates the effectiveness of each component introduced in IDPNet. First, to validate the effectiveness of the prototype adaptation methods, only the prototype adaptation constraints at the feature level (Proto-Feature) and output level (Proto-Output) are retained and compared with the baseline. Finally, both level constraints are simultaneously retained to demonstrate the effectiveness of multilevel prototype adaptation constraints. Furthermore, after retaining the prototype adaptation constraints at both levels, HFMs are added to the encoder to validate its effectiveness.

The average values of ten experiments conducted on two sets of test data are listed in Tables III and IV. From Tables III and IV, the performance of the model significantly improves when the two prototype adaptation methods are added individually to the baseline. Furthermore, incorporating both prototype adaptation constraints at multiple levels further enhances the model's effectiveness, demonstrating the benefits of modeling multilevel interclass relationships for representing domain-invariant relationships. Moreover, with the addition of the HFM, there is a further improvement in metrics such as IoU. This reflects the impact of introducing multilevel features through HFM, which aids the process of prototype adaptation and enhances the learning capability of domain-invariant features.

D. Segmentation Performance

Table V presents the test results of all networks on remote sensing data for the first quarter of 2021, with the remote sensing images captured on January 28, 2021. Among all the comparative networks, ProCA, FADA, CLAN, and SDAT exhibit better segmentation metrics. The experimental results for the first quarter of 2021 indicate that after incorporating the multilevel ID prototype adaptation constraint, IDPNet achieves the highest overall IoU score of 66.74. Furthermore, with the addition of the HFM, the precision is slightly decreased, while various metrics are significantly improved, including IoU. Compared to the best-performing network, ProCA, IDPNet demonstrates an IoU improvement of 1.96 and an F1-score improvement of 1.38. Fig. 11 illustrates the segmentation results of the IDPNet

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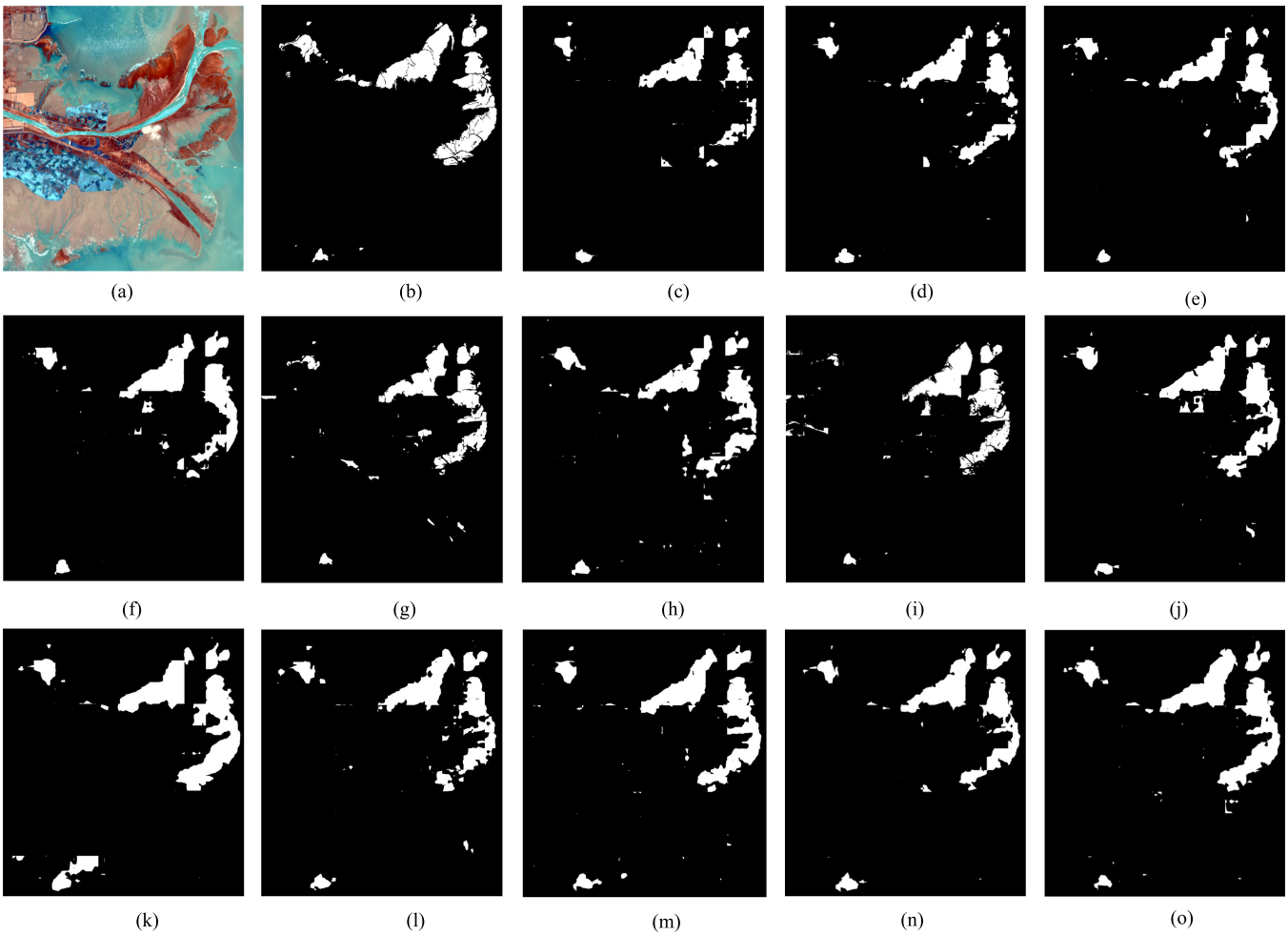


Fig. 11. Segmentation results of remote sensing data in 2021-1 by IDPNet and other networks. (•%) represents the IoU score. These results show the overall comparison of the prediction results of IDPNet and some networks for remote sensing images in 2021-1. (a) Image. (b) Label. (c) Baseline (56.2%). (d) DSCN (58.78%). (e) ADVENT (59.32%). (f) AdaptSeg (59.9%). (g) DASP (60.3%). (h) DALN (61.64%). (i) Cscada (61.74%). (j) Maxsquare (62.38%). (k) SDAT (63.0%). (l) CLAN (63.56%). (m) FADA (64.3%). (n) ProCA (65.4%). (o) Ours (67.36%).

network and the comparative networks on temporal data for the first quarter of 2021. IDPNet achieves the best segmentation results for the S.alterniflora class. The other comparative networks, to varying degrees, exhibit instances of missing S.alterniflora. Only IDPNet is capable of effectively delineating the expanded boundaries of S.alterniflora as a whole, while also distinguishing smaller regions of S.alterniflora to a significant extent. This demonstrates that the proposed ID prototype adaptation approach, in comparison to other cross-scene methods, possesses stronger capabilities in extracting domain-invariant representations, thereby achieving finer grained segmentation of S.alterniflora across scenes.

Table VI presents the test results of all networks on remote sensing data for the third quarter of 2021, with the imagery captured on June 21, 2021. It is evident that ProCA, FADA, DALN, and SDAT show relatively better segmentation metrics compared to all the comparative networks. However, for networks that performed relatively well in the first quarter of 2021, such as CLAN, there is a noticeable decrease in their ranking after the quarterly change. This is because temporal variations introduce significant changes in the features of S.alterniflora, making it challenging for some UDA methods to accurately align class features and capture the relationships between classes. Consequently, they struggle to achieve the same level of segmentation performance on test data with different temporal characteristics. In contrast, IDPNet models the interclass relationships by constructing ID prototype features, thereby reducing domain discrepancies and aligning class-level features across different temporal data. As observed in Fig. 12, even after temporal changes, IDPNet consistently achieves the best performance, highlighting the superiority of the ID prototype adaptation method in cross-scene class-level learning compared to other domain adaptation approaches and its stronger ability to extract domain-invariant features.

Furthermore, a comparison of the computational complexity is conducted for each algorithm, and the time required to run one epoch for all models is listed in Table VII. This time includes the sum of the forward and backward propagation time for the same batch of data. As listed in Table VII, adversarial learning takes a longer time due to the simultaneous training of the encoder and discriminator at different scales. In comparison, ProCA and IDPNet employ

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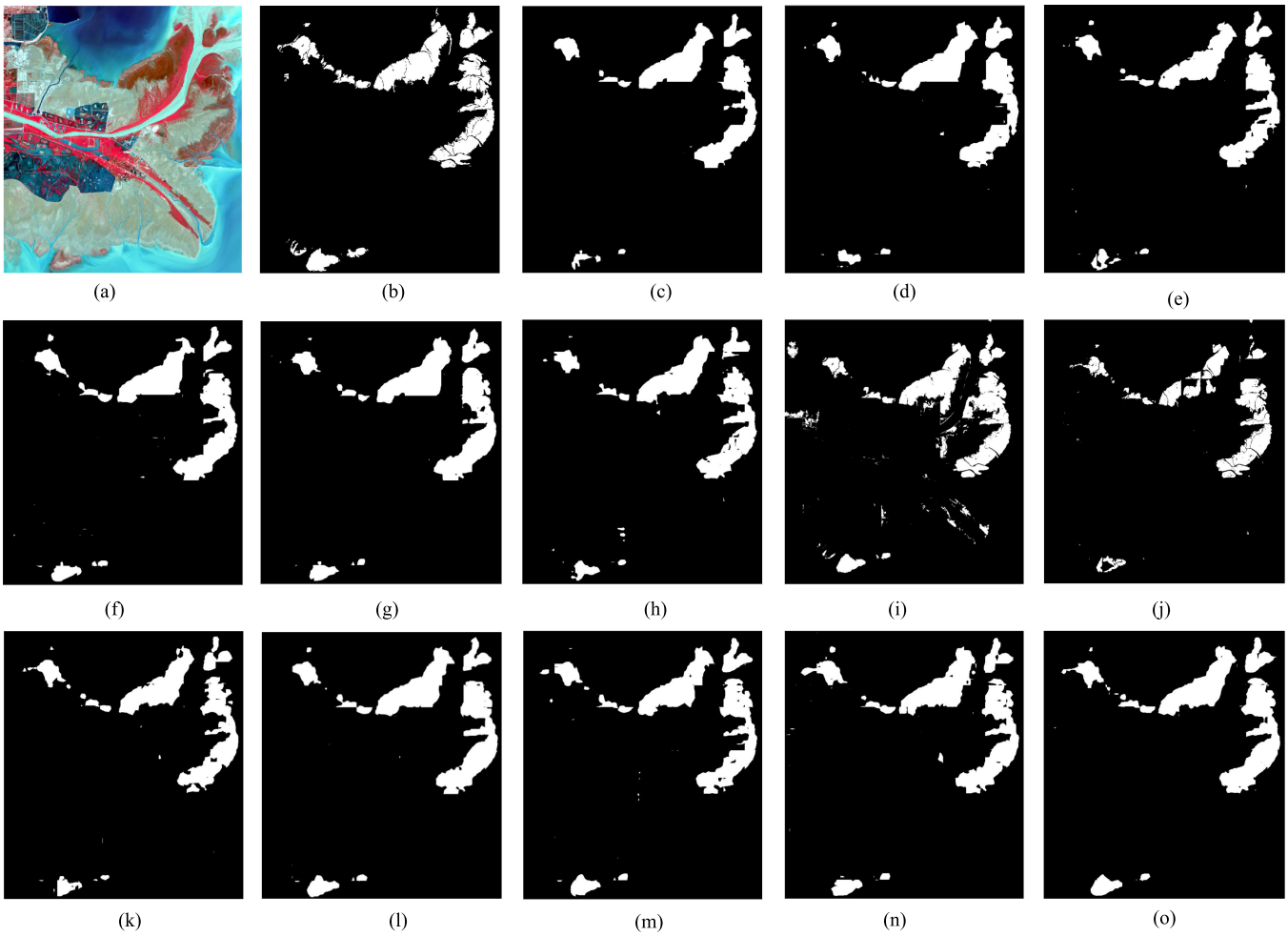


Fig. 12. Segmentation results of remote sensing data in 2021-2 by IDPNet and other networks. (•%) represents the IoU score. These results show the overall comparison of the prediction results of IDPNet and some networks for remote sensing images in 2021-2. (a) Image. (b) Label. (c) Baseline (72.92%). (d) ADVENT (73.58%). (e) AdaptSeg (74.5%). (f) DSCN (75%). (g) CLAN (75.34%). (h) Maxsquare (75.46%). (i) DASP (75.46%). (j) Cscada (76.38%). (k) DALN (76.46%). (l) SDAT (77.5%). (m) FADA (77.52%). (n) ProCA (77.76%). (o) Ours (80.18%).

TABLE VII
EXECUTION TIME (IN SECONDS) OF ONE EPOCH TRAINING OF DIFFERENT METHODS

Testing Year	Network											
	AdaptSeg	ADVENT	MaxSquare	DSCN	CLAN	FADA	DALN	DASP	Cscada	SDAT	ProCA	IDPNet
2021-1	192.26	201.30	175.74	180.66	204.69	187.20	181.34	146.73	151.45	342.44	169.83	187.87
2021-2	193.64	203.50	175.82	180.28	204.74	187.67	182.03	147.14	152.04	344.19	170.14	187.99

interclass modeling with contrastive learning, which offers better training efficiency. Although IDPNet is slower than ProCA, it leads to significant improvements in various performance metrics.

Based on the results of all sets of experiments, among all the networks, ProCA ranks second in terms of several metrics, which demonstrates that prototype adaptation based on contrastive learning is more effective than other domain adaptation methods in cross-domain detection of S.alterniflora. However, under the same encoder, IDPNet achieves higher IoU scores and other metric scores for S.alterniflora compared to ProCA. This demonstrates that the ID prototype adaptation method designed in IDPNet is superior to directly applying prototype adaptation in ProCA. It enables better modeling of interclass relationships, thereby enhancing the transferability of domain-invariant knowledge from the labeled year to the predicted year. This allows for multitemporal S.alterniflora detection across different scenes.

IV. CONCLUSION

This article presents a fine-grained IDPNet to enhance the modeling of interclass relationships in domain adaptation

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and mitigate the shift problem in cross-temporal S.alterniflora segmentation. Unlike direct prototype construction, IDPNet dynamically creates class prototypes within an ID and constructs them at multiple scales across the network's feature and output layers, boosting class interpretability in cross-temporal scenarios. Moreover, IDPNet, with its HFM, learns more robust class features, resulting in an adaptive segmentation model that generalizes effectively to the TD. Comprehensive experiments on two temporally disparate test datasets demonstrate that IDPNet significantly outperforms current advanced methods in cross-temporal S.alterniflora segmentation. At present, IDPNet may not have a significant speed advantage compared to other contrastive networks, and its training relies on TD data. This dependence is especially limiting in practical applications, without such data, the model's ability to generalize is difficult to ensure. Therefore, our future research will focus on two key issues: first, how to improve the network's operational efficiency without sacrificing accuracy. Second, we aim to enable knowledge transfer from the SD to the TD without direct data access to facilitate model generalization. This would significantly expand IDPNet's usability in scenarios where target data are unavailable.

REFERENCES

[1] J. Lin, F. Gao, X. Shi, J. Dong, and Q. Du, "SS-MAE: Spatial–spectral masked autoencoder for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023, Art. no. 5531614.

[2] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, Art. no. 5501816.

[3] E. Adam, O. Mutanga, and D. Rugege, "Multispectral and hyperspectral remote sensing for identification and mapping of wetland vegetation: A review," *Wetlands Ecology Manag.*, vol. 18, no. 3, pp. 281–296, Jun. 2010.

[4] B. Zhao et al., "Multiple attention network for spartina alterniflora segmentation using multitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023, Art. no. 5402915.

[5] K. Li, G. Zhang, X. Li, and J. Xie, "Face recognition based on improved Retinex and sparse representation," *Proc. Eng.*, vol. 15, pp. 2010–2014, Jan. 2011.

[6] J. Xu, W. An, L. Zhang, and D. Zhang, "Sparse, collaborative, or nonnegative representation: Which helps pattern classification?" *Pattern Recognit.*, vol. 88, pp. 679–688, Apr. 2019.

[7] W.-Y. Ren, M. Tang, Y. Peng, and G.-H. Li, "Semi-supervised image classification via nonnegative least-squares regression," *Multimedia Syst.*, vol. 23, no. 6, pp. 725–738, Nov. 2017.

[8] N. B. Gallagher, "Detection, classification and quantification in hyperspectral images using classical least squares models," in *Techniques and Applications of Hyperspectral Image Analysis*. 2007, pp. 181–201.

[9] Y. Gui, W. Li, X.-G. Xia, R. Tao, and A. Yue, "Infrared attention network for woodland segmentation using multispectral satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5627214.

[10] R. Kemker, C. Salvaggio, and C. Kanan, "Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 60–77, Nov. 2018.

[11] W. Li, J. Wang, Y. Gao, M. Zhang, R. Tao, and B. Zhang, "Graph-feature-enhanced selective assignment network for hyperspectral and multispectral data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5526914.

[12] L. Bruzzone and D. F. Prieto, "Unsupervised retraining of a maximum likelihood classifier for the analysis of multitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 2, pp. 456–460, 2001.

[13] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.

[14] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9646–9660, Nov. 2021.

[15] M. Zhang, X. Zhao, W. Li, Y. Zhang, R. Tao, and Q. Du, "Cross-scene joint classification of multisource data with multilevel domain adaption network," *IEEE Trans. Neural Netw. Learn. Syst.*, doi: 10.1109/TNNLS.2023.3262599.

[16] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[17] L. Li, X. Yao, X. Wang, D. Hong, G. Cheng, and J. Han, "Robust few-shot aerial image object detection via unbiased proposals filtration," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–11, 2023, Art. no. 5617011.

[18] Y. Luo, L. Zheng, T. Guan, J. Yu, and Y. Yang, "Taking a closer look at domain shift: Category-level adversaries for semantics consistent domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2502–2511.

[19] X. Yao, Q. Cao, X. Feng, G. Cheng, and J. Han, "Scale-aware detailed matching for few-shot aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, Art. no. 5611711.

[20] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, "A theory of learning from different domains," *Mach. Learn.*, vol. 79, nos. 1–2, pp. 151–175, May 2010.

[21] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. J. Smola, "A kernel method for the two-sample problem," 2008, *arXiv:0805.2368*.

[22] Y.-H. Tsai, W.-C. Hung, S. Schulter, K. Sohn, M.-H. Yang, and M. Chandraker, "Learning to adapt structured output space for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7472–7481.

[23] T.-H. Vu, H. Jain, M. Bucher, M. Cord, and P. Pérez, "ADVENT: Adversarial entropy minimization for domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2512–2521.

[24] R. Gu et al., "Contrastive semi-supervised learning for domain adaptive segmentation across similar anatomical structures," *IEEE Trans. Med. Imag.*, vol. 42, no. 1, pp. 245–256, Jan. 2023.

[25] C. Chen et al., "Progressive feature alignment for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 627–636.

[26] A. Kumar et al., "Co-regularized alignment for unsupervised domain adaptation," in *Proc. Adv. Neural Inf. Process. Syst.*, Red Hook, NY, USA: Curran Associates, 2018, pp. 9367–9378.

[27] P. Wang, J. Peng, M. Pedersoli, Y. Zhou, C. Zhang, and C. Desrosiers, "Shape-aware joint distribution alignment for cross-domain image segmentation," *IEEE Trans. Med. Imag.*, vol. 42, no. 8, pp. 2338–2347, Aug. 2023.

[28] Y. Zhu et al., "Deep subdomain adaptation network for image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 4, pp. 1713–1722, Apr. 2021.

[29] Y. Luo, P. Liu, L. Zheng, T. Guan, J. Yu, and Y. Yang, "Category-level adversarial adaptation for semantic segmentation using purified features," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 3940–3956, Aug. 2022.

[30] H. Wang, T. Shen, W. Zhang, L. Duan, and T. Mei, "Classes matter: A fine-grained adversarial approach to cross-domain semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 642–659.

[31] H. Rangwani, S. K. Aithal, M. Mishra, A. Jain, and V. B. Radhakrishnan, "A closer look at smoothness in domain adversarial training," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 18378–18399.

[32] P. Zhang, B. Zhang, T. Zhang, D. Chen, Y. Wang, and F. Wen, "Prototypical pseudo label denoising and target structure learning for domain adaptive semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Los Alamitos, CA, USA, Jun. 2021, pp. 12409–12419.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:08:21 UTC from IEEE Xplore. Restrictions apply.

[33] Z. Jiang et al., "Prototypical contrast adaptation for domain adaptive segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 36–54.

[34] Y. Dai, J. Liu, Y. Sun, Z. Tong, C. Zhang, and L.-Y. Duan, "IDM: An intermediate domain module for domain adaptive person Re-ID," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Los Alamitos, CA, USA, Oct. 2021, pp. 11844–11854.

[35] Y. Dai, Y. Sun, J. Liu, Z. Tong, Y. Yang, and L.-Y. Duan, "Bridging the source-to-target gap for cross-domain person re-identification with intermediate domains," 2022, *arXiv:2203.01682*.

[36] W. Zhu et al., "Monitoring the invasive plant *Spartina alterniflora* in Jiangsu coastal wetland using MRCNN and long-time series Landsat data," *Remote Sens.*, vol. 14, no. 11, p. 2630, May 2022.

[37] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5609413.

[38] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[39] K. Mei, C. Zhu, J. Zou, and S. Zhang, "Instance adaptive self-training for unsupervised domain adaptation," in *Computer Vision—ECCV*. Glasgow, U.K.: Springer, 2020, pp. 415–430.

[40] Z. Zheng and Y. Yang, "Rectifying pseudo label learning via uncertainty estimation for domain adaptive semantic segmentation," *Int. J. Comput. Vis.*, vol. 129, no. 4, pp. 1106–1120, Apr. 2021.

[41] W. Zeng, W. Li, and R. Tao, "IANet: Important-aware network for microscopic hyperspectral pathology image segmentation," in *Proc. 15th Int. Congr. Image Signal Process., Biomed. Eng. Informat. (CISP-BMEI)*, 2022, pp. 1–6.

[42] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.

[43] S. Mehta, M. Rastegari, L. Shapiro, and H. Hajishirzi, "ESPNetv2: A light-weight, power efficient, and general purpose convolutional neural network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9182–9192.

[44] S. Mehta, M. Rastegari, A. Caspi, L. Shapiro, and H. Hajishirzi, "Espnet: Efficient spatial pyramid of dilated convolutions for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 552–568.

[45] M. Chen, H. Xue, and D. Cai, "Domain adaptation for semantic segmentation with maximum squares loss," 2019, *arXiv:1909.13589*.

[46] L. Chen et al., "Reusing the task-specific classifier as a discriminator: Discriminator-free adversarial domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 7171–7180.

[47] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[48] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[49] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[50] L. Bottou, "Stochastic gradient descent tricks," in *Neural Networks: Tricks of the Trade*, 2nd ed. Berlin, Germany: Springer, 2012, pp. 421–436.

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